Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · Full-Stack · Python · Node.js · React · TypeScript · AWS · LLM / Agentic AI · Prototype to Production

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SUMMARY

Senior Software Engineer with 8+ years building full-stack production systems from schema design through deployment - and a demonstrated pattern of taking experimental ideas to production. Built **Leasel** from a blank page: prototyped the architecture, picked the stack, shipped an 8-agent LangGraph system with a FastAPI backend, hybrid search, document intelligence, and a CI regression harness - owned it end to end. Built **AgentProbe**, an open-source LLM evaluation toolkit with a 15-endpoint FastAPI backend, multi-provider routing (OpenAI, Anthropic, Azure OpenAI, Groq, Ollama), and CI-integrated golden dataset testing - designed to give AI engineers fast, real performance data to inform decisions. At Entrata (8 years, 100K+ monthly active users), owned backend APIs, data pipelines, async distributed processing, AWS infrastructure, and full-stack modules across a multi-tenant SaaS platform in a regulated environment. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

Python · Node.js · TypeScript | React · JavaScript | REST APIs · OpenAPI | FastAPI · Django | LLMs · Embeddings · Agents | LangGraph · LangChain | RAG · Hybrid Search | AWS (Lambda · S3 · RDS · API GW) | PostgreSQL · NoSQL · Redis | Docker · Kubernetes | CI/CD · GitHub Actions | Observability · Structured Logging | Schema Design · Data Modeling | Authentication · Authorization | Multi-Tenant Architecture | Async Processing · RabbitMQ | Java · PHP · Kotlin · Scala | Agile · Scrum · Mentorship

EXPERIENCE

AI/ML Engineer (Volunteer)

Sep 2026 - Present

Blue Horse Digital — Virginia, USA

- Building ML models, data pipelines, and full-stack API layers connecting AI components to backend services and reporting systems; debugging and optimizing data-driven features across the stack.

AI/ML Engineer Intern

May 2025 - Aug 2025

Magizhs Tech Solution

- Prototyped and shipped an agentic mission-planning system (LangGraph) - natural-language input to structured flight plans with RAG over regulatory documents, validation guardrails, and conditional routing to export or rejection; deployed MCP server connecting agentic outputs to enterprise tooling.
- Engineered full document ingestion, chunking, retrieval, and FastAPI streaming endpoint pipeline from regulatory corpus to real-time operator interface.

Senior Software Engineer

Sep 2018 - Aug 2024

Entrata - PropTech / Real Estate SaaS (100K+ monthly active users, multi-tenant)

- **Shipped 20+ full-stack production modules** end-to-end (Python, PHP, PostgreSQL, JavaScript, React, RabbitMQ) across 120+ Agile sprints - database schema through API design through UI components through deployment pipelines.
- **Owned backend APIs and data models** across multi-tenant SaaS serving 100K+ users - REST API design, schema migrations, query performance optimization (200+ queries, 30% gain), and third-party integrations with defined API contracts and authentication/authorization patterns.
- **Built and operated distributed async processing** (RabbitMQ) for tenant record updates - retry logic, idempotency, production incident ownership; owned annual data pipeline (few thousand federal rows → ~180,000 domain records) powering eligibility, rent, and compliance calculations for 7 years, zero violations.
- **AWS infrastructure**: led S3 document migration (PHI-encrypted, 100% coverage, zero incidents), EC2/RDS production environments, and Python/Bash automation pipelines eliminating 80% of manual operational workload.
- Led microservices redesign (Docker) - 50% scalability improvement, 30% deployment time reduction; mentored 5+ engineers; maintained bug count below 10 for 6+ consecutive months.

Associate Software Engineer

Jul 2017 - Sep 2018

Entrata India, Pune

- Built backend modules (Python, PHP, Java, PostgreSQL) including large-scale data imports (1.6M records), document management, and compliance workflows; led AWS S3 cloud migration from legacy storage.

PROJECTS

Leasel - Agentic AI Platform (Prototype to Production)

2026 | Python, LangGraph (8-agent), FastAPI, SQLite, CI/CD | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Took the problem from blank page to deployed system: chose the architecture (LangGraph StateGraph), designed the schema, built three production surfaces sharing one SQLite knowledge repository - document intelligence (OCR, hybrid search, citation grounding), agentic investigation (8-agent pipeline with HITL escalation), and tenant self-service with a hard eligibility boundary.
- CI regression gate with LLM-as-judge scoring; 93.6% precision and 87.9% F1 across 1,240 test cases (10 compliance rules, 5 US states + federal).

AgentProbe - Open-Source LLM Evaluation Toolkit

2025 | Python, FastAPI, LangGraph, CI/CD | github.com/venkatalakshmiparimala/AgentProbe

- 15-endpoint FastAPI backend; 5-metric eval engine; multi-provider LLM routing (OpenAI, Anthropic, Azure OpenAI, Groq, Ollama); CI-integrated regression harness with golden datasets and structured logging - gives engineers real performance data to make AI integration decisions, not just vibes.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia | GPA 3.87 May 2026

B.Tech Computer Science & Engineering — JNTU Anantapur 2017